深圳市创力信电子有限公司

PRODUCT SPECIFICATIONS

For Customer:_____ ☐ : APPROVAL FOR SPECIFICATION

Customer Model No._____ ☐ : APPROVAL FOR SAMPLE

Module No.: SZCLX070H-500-IPS-G-RTP Date : 2024-03-21

Version :A

1. Table of Contents

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For Customer's Acceptance:

Approved By	Comment

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2. Revision Record

Date	Rev.No.	Page	Revision Items	Prepared
2024.03.21	A		The first release	

3. General Specifications

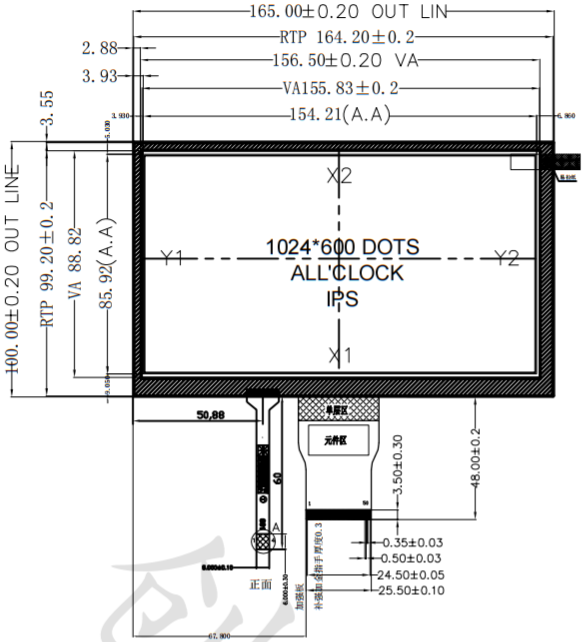
SZCLX070H-500-IPS-G-RTP is a TFT-LCDmodule.ItiscomposedofaTFT-LCDpanel, driver IC, FPC, a back light unit. The 7.0′′ display area contains 1024 x 600 pixels and can display up to 16.7M colors. This product accords with RoHS environmental criterion.

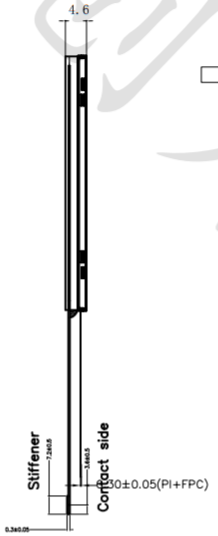
Item	Contents	Unit	Note
LCD Type	Normally Black,Transmissive	-	
Display color	16.7M		1
Viewing Direction	ALL	O'Clock	
Operating temperature	-20~+70	℃	
Storage temperature	-30~+80	℃	
Module size	165.00(W)×100.00(H)×3.50(T)	mm	2
Active Area(W×H)	154.2144(W)×85.92(H)	mm	
Number of Dots	1024×RGB×600	dots	
Backlight	27-LEDs (white)	pcs	
Interface	RGB Interface	-	
Driver IC	HX8282-A11+HX8696-A		

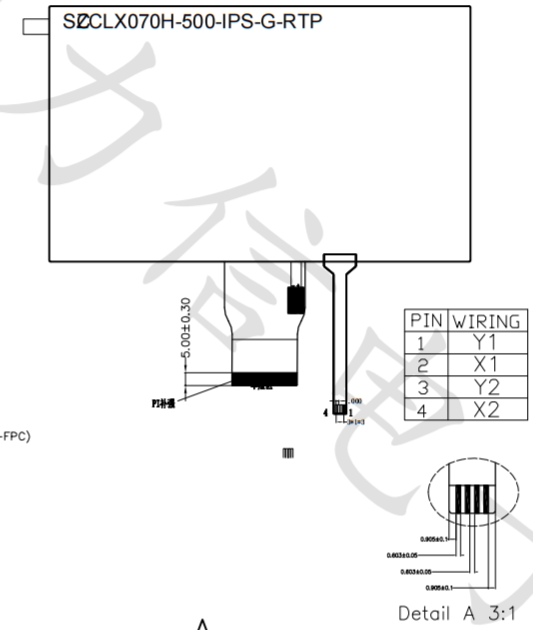
Note 1: Color tune is slightly changed by temperature and driving voltage.

Note 2: Without FPC and Solder.

REV	DESCRIPTION	DATE
A	初版	2023-06-28

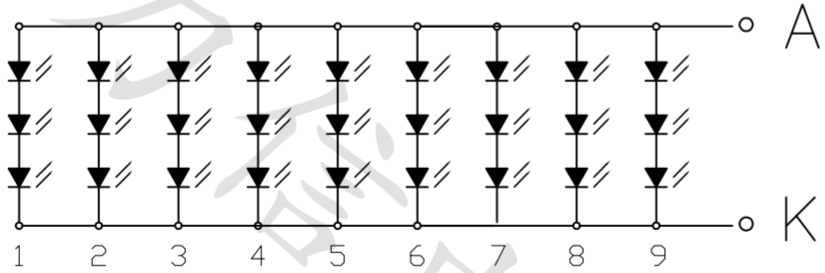






PIN	WIRING
1	Y1
2	X1
3	Y2
4	X2

PIN	SYMBOL	PIN	SYMBOL
1	VLED+	26	G1
2	VLED+	27	G0
3	VLED-	28	R7
4	VLED-	29	R6
5	GND	30	R5
6	VCOM	31	R4
7	DVDD	32	R3
8	MODE	33	R2
9	DE	34	R1
10	VSYNC	35	R0
11	HSYNC	36	GND
12	B7	37	DCLK
13	B6	38	GND
14	B5	39	SHLR
15	B4	40	UPDN
16	B3	41	VGH
17	B2	42	VGL
18	B1	43	AVDD
19	B0	44	RESET
20	G7	45	NC
21	G6	46	VCOM
22	G5	47	DITHB
23	G4	48	GND
24	G3	49	NC
25	G2	50	NC



LED 电路图

Vf 9.0-10.8V

IF=20*9=180mA(定电流测试)

Specification:

1). Display mode: 16.7M, TFT/Transmissive
2). Operating temp.: -30°C~+80°C
Storage temp.: -35°C~+85°C
3). Backlight: 27 chip White LED
4). Dimensions with mark "*" are important, with mark "()" are referenced

深圳市创力信电子有限公司
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TITLE:	ASSEMBLY	DO NOT SCALE THIS DRAWING, GENERAL TOLERANCE:±0.2			
MODULE NO.:	SZCLX070H-500-IPS-G-RTP			REV	A
DRAWN BY:		DATE:	2024-03-21	PROJECTION:	
CHECKED BY:		DATE:	2024-03-21	SCALE:N.T.S	UNIT: mm
APPROVED BY:		DATE:	2024-03-21	SHEET: 1 OF 1	

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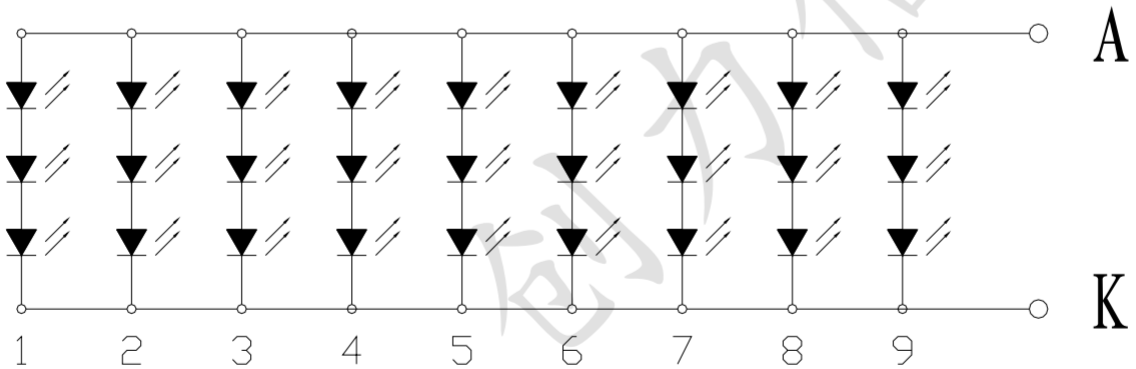
5.ElectricalSpecificationsandTimingCharacteristics

5.1Electricalcharacteristics(V ss=0V ,Ta=25℃)

Item	Symbol	Min.	Typ	Max.	Unit
Digital Supply Voltage	**VDD**	3.0	3.3	3.6	**V**
Analog Supply Voltage	**AVDD**	8.9	9.0	9.1	**V**
TFT Gate ON Voltage	**VGH**	17.0	18.0	19.0	**V**
TFT Gate OFF Voltage	**VGL**	-6.5	-6.0	-5.5	**V**
TFT Common Electrode Voltage	**VCOM**	3.0	3.15	3.3	**V**

5.2LEDbacklightspecification(V ss=0V ,Ta=25℃)

Item	Symbol	Condition	Min	Typ	Max	Unit	Note
Supply voltage	-	-	9.0	9.6	10.8	V	1
Supply current	I_f	-	-	180	-	mA	2



Note:

1: $V_{LED}=V_{LED}(+)-V_{LED}(-)$.

2:The current of LED is 20mA.
A LED drive in constant current mode is recommended.

5.3Interfacesignals

Pin NO.	SYMBOL	I/O	DESCRIPTION	Remark
1	VLED+	P	Power for LED backlight (Anode)	Note 8
2	VLED+	P	Power for LED backlight (Anode)	Note 8
3	VLED-	P	Power for LED backlight (Cathode)	Note 8
4	VLED-	P	Power for LED backlight (Cathode)	Note 8
5	GND	P	Power ground	
6	VCOM	I	Common Voltage	
7	DVDD	P	Digital Power	
8	MODE	I	DE/SYNC mode select.	Note 1
9	DE	I	Data Enable signal	
10	VSYNC	I	Vertical sync input	
11	HSYNC	I	Horizontal sync input	
12	B7	I	Blue data (MSB)	
13	B6	I	Blue data	
14	B5	I	Blue data	
15	B4	I	Blue data	
16	B3	I	Blue data	
17	B2	I	Blue data	
18	B1	I	Blue data	Note 2
19	B0	I	Blue data (LSB)	Note 2
20	G7	I	Green data (MSB)	
21	G6	I	Green data	
22	G5	I	Green data	
23	G4	I	Green data	
24	G3	I	Green data	
25	G2	I	Green data	
26	G1	I	Green data	Note 2
27	G0	I	Green data (LSB)	Note 2

28	R7	I	Red data (MSB)	
29	R6	I	Red data	
30	R5	I	Red data	
31	R4	I	Red data	
32	R3	I	Red data	
33	R2	I	Red data	
34	R1	I	Red data	Note 2
35	R0	I	Red data (LSB)	Note 2
36	GND	P	Power ground	
37	DCLK	I	Clock input	Note 3
38	GND	P	Power ground	
39	SHLR	I	Left / Right Selection	Note 4,5
40	UPDN	I	Up / Down Selection	Note 4,5
41	VGH	P	Gate ON Voltage	
42	VGL	P	Gate OFF Voltage	
43	AVDD	P	Power for Analog Circuit	
44	RESET	I	Global reset pin	Note 6
45	NC	-	Not connection	
46	VCOM	I	Common Voltage	
47	DITH	I	Dithering function enable control	Note 7
48	GND	P	Power ground	
49	NC	-	Not connection	
50	NC	-	Not connection	

I: input, O: output, P: Power

Note 1: DE / SYNC mode select. Normally pull high.
When MODE=H, DE mode.
When MODE=L, SYNC mode.

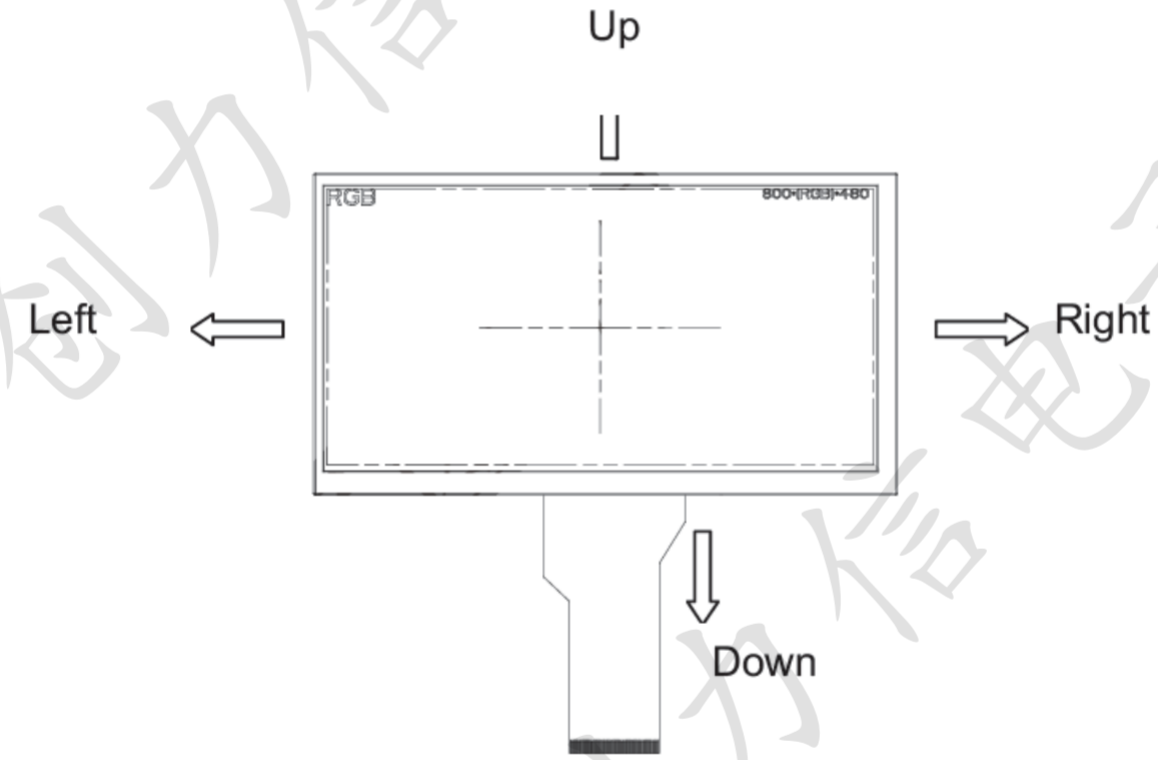
Note 2: When input 18 bits RGB data,the two low bits of R,G and B data must be grounded

Note 2: Data shall be latched at the falling edge of DCLK.

Note 4: Selection of scanning mode

Setting of scan control input		Scanning direction
U/D	L/R	
GND	DV_{DD}	Up to down, left to right
DV_{DD}	GND	Down to up, right to left
GND	GND	Up to down, right to left
DV_{DD}	DV_{DD}	Down to up, left to right

Note 5: Definition of scanning direction.
Refer to the figure as below:



Note 6: Global reset pin. Active low to enter reset state. Suggest to connect with an RC reset circuit for stability. Normally pull high.

Note 7:Dithering function enable control. Normally pull low
DITHER = “1”, Enable internal dithering function
DITHER = “0”, Disable internal dithering function

Note 8: Reserve for LED power input.

6.OpticalCharacteristics

Item	Symbol		Condition	Min.	Typ.	Max.	Unit	Note
Brightness	Bp		θ=0° Φ=0°	-	500	-	Cd/m²	1
Uniformity	⊿Bp			75	80	-	%	1,2
Viewing Angle	3:00		Cr≥10	80	85	-	Deg	3
	6:00			80	85	-		
	9:00			80	85	-		
	12:00			80	85	-		
Contrast Ratio	Cr		θ=0° Φ=0°	500	800		-	4
Response Time	T_r+T_f				25	40	ms	5
Color of CIE Coordinate	W	x	θ=0° Φ=0°	TYP -0.05	0.303	TYP +0.05	-	1,6
		y			0.333		-	
	R	x			-		-	
		y			-		-	
	G	x			-		-	
		y			-		-	
	B	x			-		-	
		y					-	

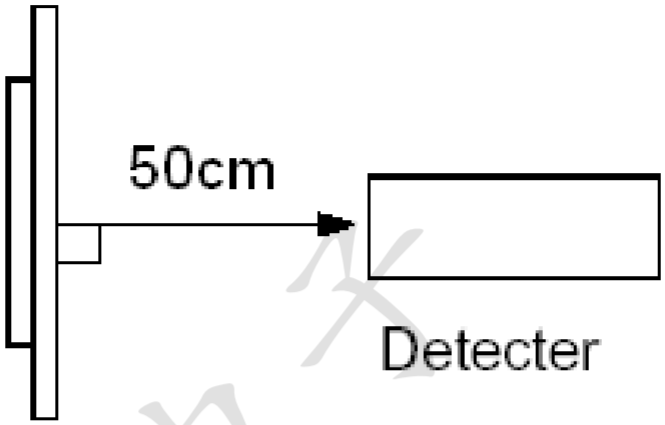
Note：The parameter is slightly changed by temperature, driving voltage and materiel

Note 1: The data are measured after LEDs are turned on for 5 minutes. LCM displays full white. The brightness is the average value of 9 measured spots. Measurement equipment PR-705 (Φ8mm)

Measuring condition:

- Measuring surroundings: Dark room.
- Measuring temperature: Ta=25℃.
- Adjust operating voltage to get optimum contrast at the center of the display.

Measured value at the center point of LCD panel after more than 5 minutes while backlight turning on.

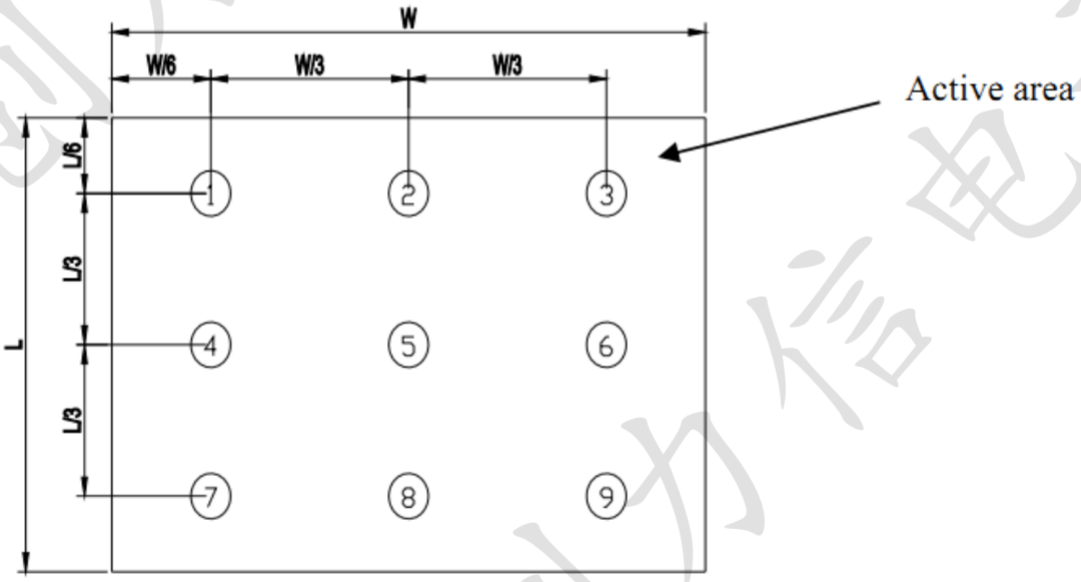


Note 2: The luminance uniformity is calculated by using following formula.

⊿Bp = Bp (Min.) / Bp (Max.)×100 (%)

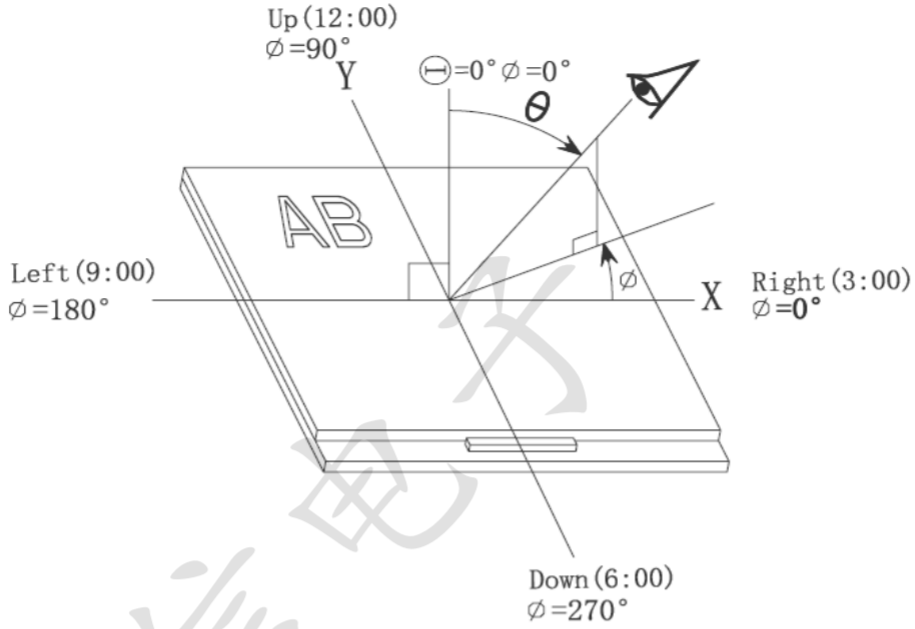
Bp (Max.) = Maximum brightness in 9 measured spots

Bp (Min.) = Minimum brightness in 9 measured spots.

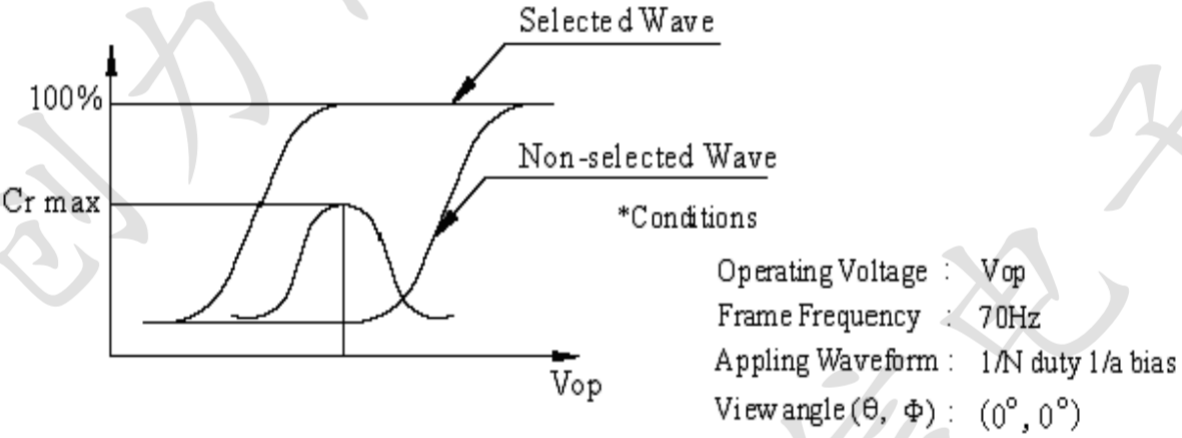


Note 3: The definition of viewing angle:

Refer to the graph below marked by θ and Φ



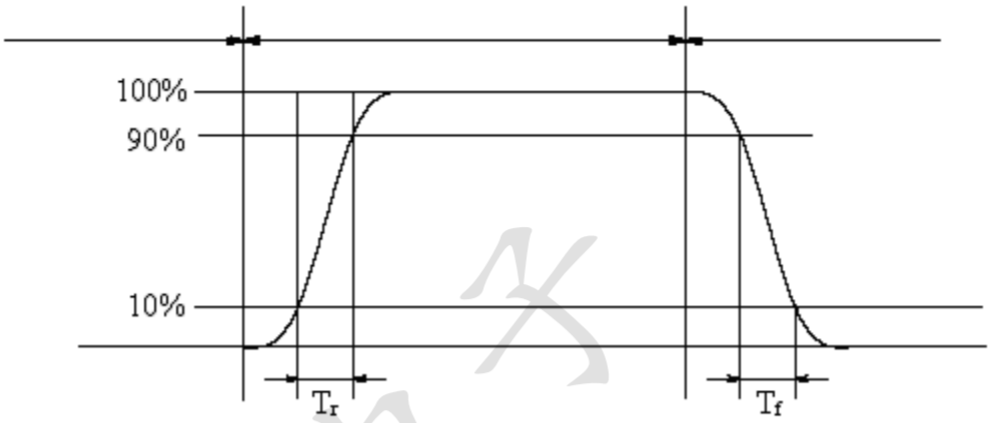
Note 4: Definition of contrast ratio.(Test LCD using DMS501)



$$Contrast\ ratio(Cr) = \frac{Brightness\ of\ selected\ dots}{Brightness\ of\ non-selected\ dots}$$

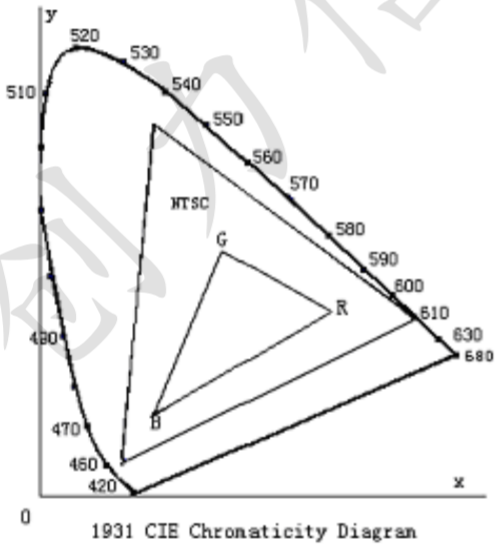
Note 5: Definition of Response time. (Test LCD using DMS501):

The output signals of photo detector are measured when the input signals are changed from “black” to “white”(falling time) and from “white” to “black”(rising time), respectively. The response time is defined as the time interval between the 10% and 90% of amplitudes.Refer to figure as below.



The definition of response time

Note 6: Definition of Color of CIE Coordinate and NTSC Ratio.

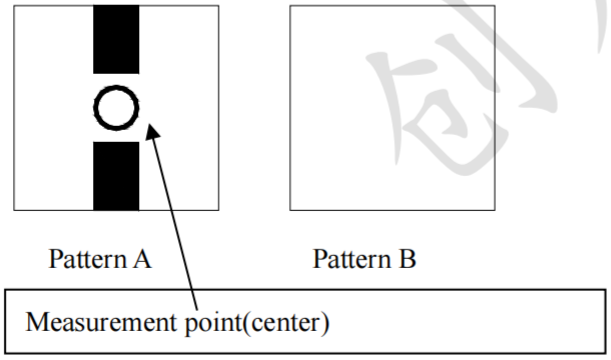


Color gamut:

$$S = \frac{\text{area of RGB triangle}}{\text{area of NTSC triangle}} \times 100\%$$

Note 7: Definition of cross talk.

Cross talk ratio(%)=|pattern A Brightness-pattern B Brightness|/pattern A Brightness*100



Electric volume value=3F+/-3Hex

7.ReliabilityTestItemsandCriteria

No	Test Item	Test condition	Criterion
1	High Temperature Storage	80℃±2℃ 96H Restore 2H at 25℃ Power off	1. After testing, cosmetic and electrical defects should not happen. 2. Total current consumption should not be more than twice of initial value.
2	Low Temperature Storage	-30℃±2℃ 96H Restore 2H at 25℃ Power off	
3	High Temperature Operation	70℃±2℃ 96H Restore 2H at 25℃ Power on	
4	Low Temperature Operation	-20℃±2℃ 96H Restore 4H at 25℃ Power on	
5	High Temperature/Humidity Operation	50℃±2℃ 90%RH 96H Power on	
6	Temperature Cycle(Storage)	–20℃←-25℃------→70℃ 30min 5min 30min after 5 cycle, Restore 2H at 25℃ Power off	
7	Vibration Test	10Hz~150Hz, 100m/s^2, 120min	Not allowed cosmetic and electrical defects.
8	Shock Test	Half- sine wave,300m/s^2,11ms	
9	ESD Test	Air discharge:+/-8KV, Contact discharge:4KV	

Note: Operation: Supply 3.3V for logic system.

The inspection terms after reliability test, as below

ITEM	Inspection
Contrast	CR>50%
IDD	IDD<200%
Brightness	Brightness>60%
Color Tone	Color Tone+/-0,05

8.Qualitylevel

8.1Classificationofdefects

Major defects (MA): A major defect refers to a defect that may substantially

degrade usability for product applications, including all functional defects(such as no display, abnormal display, open or missing segment, short circuit, missing component), outline dimension beyond the drawing, progressive defects and those affecting reliability.

Minor defects (MI): A minor defect refers to a defect which is not considered to be able to substantially degrade the product application or a defect that deviates from existing standards almost unrelated to the effective use of the product or its operation, such as black spot, white spot, bright spot, pinhole, black line, white line, contrast variation, glass defect, polarizer defect, etc.

8.2Definitionofinspectionrange

For dot defect of TFT LCD which is not smaller than 3 inches, dividing three areas to make a judgment (according to figure 1).

A area : center of viewing area

B area : periphery of viewing area

C area : Outside viewing area

For other defects, dividing two areas to make a judgment (according figure 2).

A zone : Inside Viewing area

B zone : Outside Viewing area

X1(A.A~V.A): 2mm X2(A.A~V.A): 2mm

Y1(A.A~V.A): 2mm Y2(A.A~V.A): 2mm

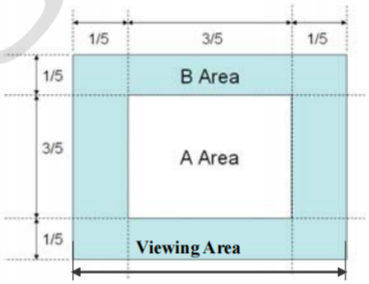


Figure 1

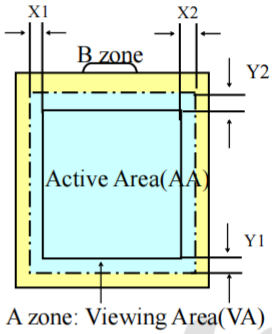


Figure 2

8.3Inspectionitemsandgeneralnotes

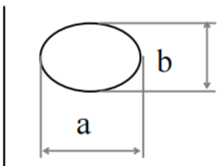
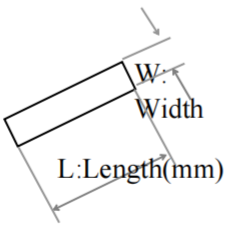
General notes	1.Should any defects which are not specified in this standard happen, additional standard shall be determined by mutual agreement between customer and C H U A N G L I X I N . 2.Viewing area should be the area which C H U A N G L I X I N guarantees. 3.Limit sample should be prior to this Inspection standard. 4.Viewing judgment should be under static pattern. 5.Inspection conditions Inspection distance: 250 mm (from the sample) Temperature : 25±5 ºC Inspection angle : 45 degrees in 6 o'clock direction (all defects in viewing area should be inspected from this direction)	
Inspection items	Pinhole, Bright spot, Black spot, White spot, Black line, White Line, Foreign particle, Bubble	The color of a small area is different from the remainder. The phenomenon doesn't change with voltage
	Contrast variation	The color of a small area is different from the remainder. The phenomenon changes with voltage
	Polarizer defect	Scratch, Dirt, Particle, Bubble on polarizer or between polarizer and glass
	Dot defect (TFT LCD)	The pixel appears bright or dark abnormally when display
	Functional defect	No display, Abnormal display, Open or missing segment, Short circuit, False viewing direction

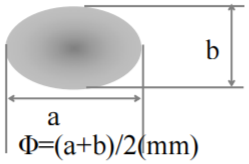
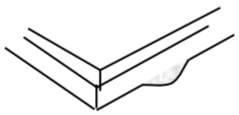
	Glass defect	Glass crack, Shaved corner of glass, Surplus glass
	PCB defect	Components assembly defect

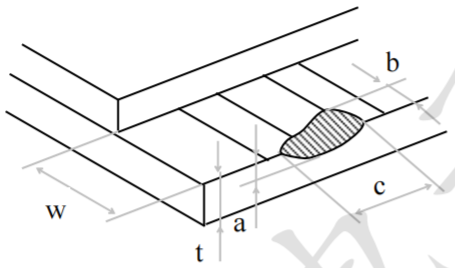
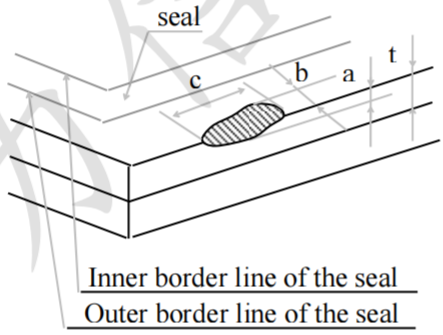
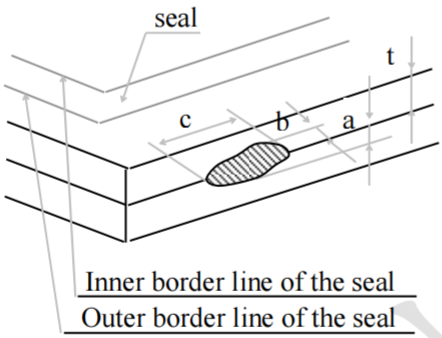
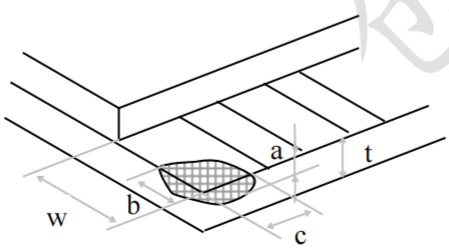
8.4OutgoingInspectionlevel

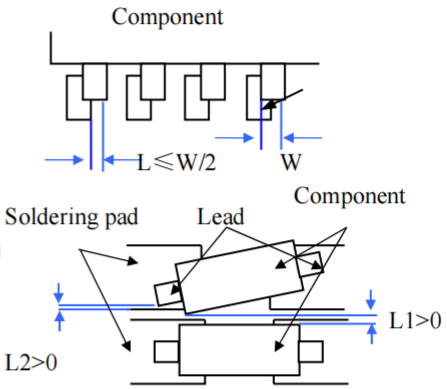
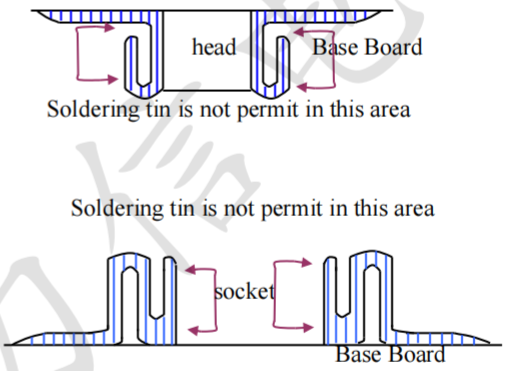
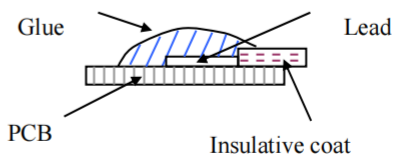
Outgoing Inspection standard	Inspection conditions	Inspection				
		Min.	Max.	Unit	IL	AQL
Major Defects	See 8.3 general notes	See 8.5			II	0.065
Minor Defects	See 8.3 general notes	See 8.5			II	0.065
Note：Sampling standard conforms to GB2828						

8.5InspectionItemsandCriteria

Inspection items			Judgment standard			
			Category		Acceptable number	
					A zone	B zone
1	Black spot, White spot, Pinhole, Foreign Particle, Particle in or on glass, Scratch on glass	 Φ=(a+b)/2(mm (a/b<2.5)	A	Φ<=0.20	Neglected	Neglected
			B	0.20<Φ<=0.25	3	Neglected
			C	0.25<Φ<=0.3	2	Neglected
			D	0.3<Φ<=0.4	1	3
			E	0.4<Φ<=0.5	0	2
			Total defective point(B,C)		1	-
2	Black line, White line, and Particle Between Polarizer and glass, Scratch on glass	 L/W>=2.5	A	W<=0.03	Neglected	Neglected
			B	0.03<W<=0.05 L<=3.0	3	Neglected
			C	0.05<W<=0.1 L<=3.0	2	Neglected
			D	0.05<W<=0.1 L<=4.0	1	3
			E	W>0.1 L>4.0	0	2
			Total defective point(B,C)		1	-

<table>
<tr><td>3</td><td colspan="2">Bright spot</td><td colspan="2">any size</td><td>none</td><td>none</td></tr>
<tr><td rowspan="5">4</td><td rowspan="5">Contrast variation</td><td rowspan="5"> 
</td><td>A</td><td>Φ<0.2</td><td>Neglected</td><td rowspan="4">Neglected</td></tr>
<tr><td>B</td><td>0.2<Φ<=0.3</td><td>2</td></tr>
<tr><td>C</td><td>0.3<Φ<=0.4</td><td>1</td></tr>
<tr><td>D</td><td>0.4<Φ</td><td>0</td></tr>
<tr><td colspan="2">Total defective point(B,C)</td><td>3</td><td></td></tr>
<tr><td>5</td><td colspan="2">Bubble inside cell</td><td colspan="2">any size</td><td>none</td><td>none</td></tr>
<tr><td rowspan="4">6</td><td rowspan="4">Polarizer defect (if Polarizer is used)</td><td>Scratch ,damage on polarizer, Particle on polarizer or between polarizer and glass.</td><td colspan="4">Refer to item 1 and item 2.</td></tr>
<tr><td rowspan="3">Bubble, dent and convex</td><td>A</td><td>Φ<=0.1</td><td>Neglected</td><td>Neglected</td></tr>
<tr><td>B</td><td>0.1 <Φ<=0.2</td><td>2</td><td>Neglected</td></tr>
<tr><td>C</td><td>0.2 <Φ<=0.3</td><td>1</td><td>2</td></tr>
<tr><td rowspan="2">7</td><td rowspan="2">Surplus glass</td><td>Stage surplus glass
 
</td><td colspan="4">B<=0.3mm</td></tr>
<tr><td>Surrounding surplus glass
  </td><td colspan="4">Should not influence outline dimension and assembling.</td></tr>
<tr><td>8</td><td colspan="2">Open segment or open common</td><td colspan="4">Not permitted</td></tr>
<tr><td>9</td><td colspan="2">Short circuit</td><td colspan="4">Not permitted</td></tr>
<tr><td>10</td><td colspan="2">False viewing direction</td><td colspan="4">Not permitted</td></tr>
<tr><td>11</td><td colspan="2">Contrast ratio uneven</td><td colspan="4">According to the limit specimen</td></tr>
<tr><td>12</td><td colspan="2">Crosstalk</td><td colspan="4">According to the limit specimen</td></tr>
<tr><td>13</td><td colspan="2">Black /White spot(display)</td><td colspan="4">Refer to item 1</td></tr>
<tr><td>14</td><td colspan="2">Black /White line(display)</td><td colspan="4">Refer to item 2</td></tr>
</table>

Inspection items			Judgment standard		
			Category(application: B zone)		Acceptable number
15	Glass defect crack	i) The front of lead terminals 	A	a≤ t, b≤1/5W, c≤3mm	Max.3 defects allowed
			B	Crack at two sides of lead terminals should not cover patterns and alignment mark	
		ii) Surrounding crack–non-contact side 	b < Inner borderline of the seal		
		iii) Surrounding crack– contact side 	b < Outer borderline of the seal		
		iv) Corner 	A	a <= t, b <= 3.0, c <= 3.0	
			B	Glass crack should not cover patterns u and alignment mark and patterns.	

Inspection items			Judgment standard Category(application: B zone)
16	PCB defect	Component soldering: No cold soldering、short、open circuit、burr、tin ball The flat encapsulation component position deviation must be less than 1/3 width of the pin (Pic.1)； the sheet component deviation: Pin deviates from the pad and contact with the near components is not permitted（Pic.2）	
		lead defect: The lead lack must be less than 1/3 of its width; The lead burr must be less than 1/3 of the seam; Impurities connect with the near leads is not permitted	
		Connector soldering: Soldering tin is at contact position of the plug and socket is not permitted No foundation is scald Serious cave distortion on plug and socket contact pin is not permitted	
		Glue on root of the speaker receiver and motor lead: The insulative coat of the lead must join into the PCB; the protected glue must envelop to the insulative coat.	

9.PrecautionsforUseofLCDModules

9.1HandlingPrecautions

9.1.1Thedisplaypanelismadeofglass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

9.1.2Ifthedisplaypanelisdamagedandtheliquidcrystalsubstanceinsideit leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

9.1.3Donotapplyexcessiveforcetothedi splay surface or the adjoining areas since this may cause the color tone to vary.

9.1.4ThepolarizercoveringthedisplaysurfaceoftheLCDmoduleissoftand easily scratched. Handle this polarizer carefully.

9.1.5Ifthedisplaysurfaceiscontam inated, breathe on the surface and gently wipe it with a soft dry cloth. If still not completely clear, moisten cloth with one of the following solvents:

— Isopropyl alcohol

— Ethyl alcohol

Solvents other than those mentioned above may damage the polarizer. Especially, do not use the following:

— Water

— Ketone

— Aromatic solvents

9.1.6DonotattempttodisassembletheLCDModule.

9.1.7Ifthelogiccircuitpowerisoff,donotapplytheinputsignals.

9.1.8Topreventdestructionoftheelemen ts by static electricity, be careful to maintain an optimum work environment.

a. Be sure to ground the body when handling the LCD Modules.

b. Tools required for assembly, such as soldering irons, must be properly ground.

c. To reduce the amount of static electricity generated, do not conduct

assembly and other work under dry conditions.

d. The LCD Module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

9.2Storageprecautions

9.2.1WhenstoringtheLCDmodules,avoid exposure to direct sunlight or to the light of fluorescent lamps.

9.2.2TheLCDmodulesshouldbestored under the storage temperature range. If the LCD modules will be stored for a long time, the recommend condition is:

Temperature : 0℃ ～ 40℃

Relatively humidity: ≤80%

9.2.3TheLCDmodulesshouldbestor ed in the room without acid, alkali and harmful gas.

9.3TheLCDmodulesshouldbenofallingandviolentshockingduring transportation, and also should avoid excessive press, water, damp and sunshine.